

Task Section 4.5

353

$$7) y = \frac{1}{5}x^5 - \frac{8}{3}x^3 + 76x$$

clmt x

$$\begin{aligned} \frac{1}{5}x^5 - \frac{8}{3}x^3 + 76x &= 0 \\ x(\frac{1}{5}x^4 - \frac{8}{3}x^2 + 76) &= 0 \\ x &= 0 \end{aligned}$$

$$\begin{aligned} y' &= x^4 - 8x^2 + 76 \\ x^4 - 8x^2 + 76 &= 0 \\ (x^2 - 4)^2 &= 0 \\ x^2 &= 4 \\ x &= \pm \sqrt{4} \\ x &= -2, 2 \end{aligned}$$

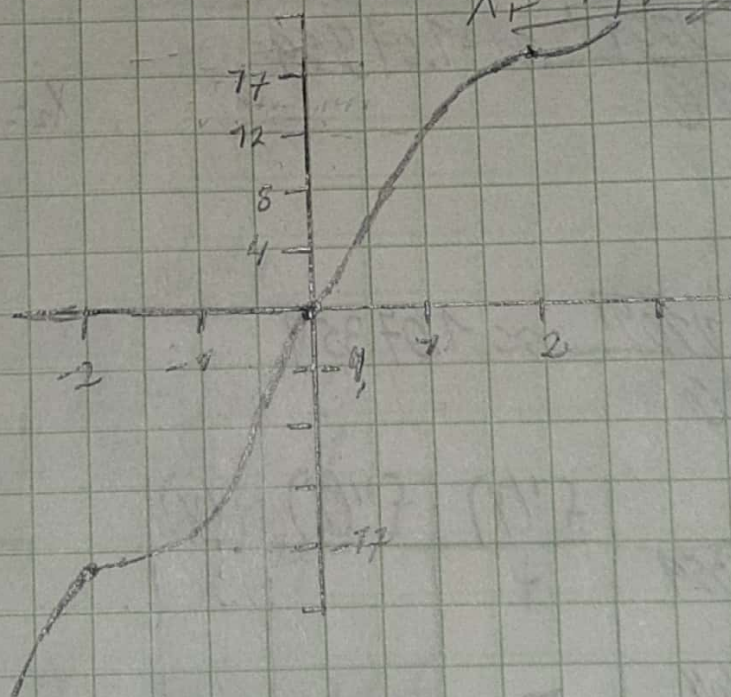
	$f'(x)$	$f''(x)$	$f(x)$
$-\infty, -2$	+	-	
-2			-77.06
$-2, 0$	+	+	
0			0
$0, 2$	+	-	
2			77.06
$2, \infty$	+	+	

clmt y

$$\begin{aligned} f(0) &= \frac{1}{5}(0)^5 - \frac{8}{3}(0)^3 + 76(0) \\ y &= 0 \end{aligned}$$

$$\begin{aligned} y'' &= 4x^3 - 76x \\ 4x^3 - 76x &= 0 \\ 4x(x^2 - 19) &= 0 \\ x &= 0 \end{aligned}$$

$$\begin{aligned} x^2 - 19 &= 0 \\ x &= \pm \sqrt{19} \\ x &= -4.36, 4.36 \end{aligned}$$



$$71) y = \frac{x - x^2}{2 - 3x + x^2}$$

1297

clmt x

$$\frac{x - x^2}{2 - 3x + x^2} = 0 \quad \frac{-x}{x-2} = 0$$

$$\frac{-x(-1+x)}{(x-1)(x-2)} = 0 \quad x=0$$

$$y' = \frac{-x}{x-2}$$

$$y' = -\frac{1(x-2) - x(1)}{(x-2)^2} = \frac{2}{(x-2)^2}$$

$$\frac{2}{(x-2)^2} = 0$$

$$2 = 0$$

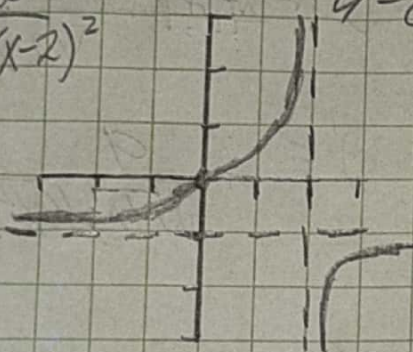
clmt y

$$f(0) = y = 0$$

$$y'' = \frac{-2 \cdot 2(x-2)}{(x-2)^{3/2}} = \frac{4}{(x-2)^3}$$

$$\frac{4}{(x-2)^3} = 0$$

$$4 = 0$$



$$72) y = \frac{x^3}{x^3 + 1}$$

clmt x

$$\frac{x^3}{x^3 + 1} = 0$$

$$x^3 = 0$$

$$x = 0$$

$$y' = \frac{3x^2(x^3 + 1) - x^3 \cdot 3x^2}{(x^3 + 1)^2}$$

$$y' = \frac{3x^2}{(x^3 + 1)^2} = 0$$

$$3x^2 = 0 \quad x = 0$$

clmt y

$$f(0) = y = 0$$

$$y'' = \frac{10x(x^3 + 1)^2 - 3x^2(2x^3 + 1)3x^2}{(x^3 + 1)^4}$$

$$y'' = \frac{-12x^4 + 6x}{(x^3 + 1)^3} = 0$$

$$12x^4 + 6x = 0$$

$$6x(2x^3 + 1) = 0$$

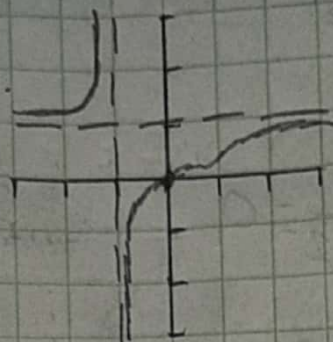
$$2x^3 + 1 = 0$$

$$x = \sqrt[3]{-\frac{1}{2}} = -0.79$$

$$x = 0$$

Scribe

	$f'(x)$	$f''(x)$	$f(x)$
$-\infty, -0.79$	+	+	
$-0.79, \infty$	+	-	7.69



23) $y = \sqrt{x^2 + x - 2}$

derivative

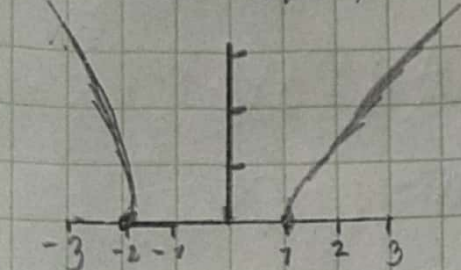
$$\sqrt{x^2 + x - 2} = 0$$

$$x^2 + x - 2 = 0$$

$$x(x+2) - (x+2) = 0$$

$$(x+2)(x-1) = 0$$

$$x = -2 \quad x = 1$$



$$y' = \frac{1}{2\sqrt{x^2 + x - 2}} \cdot (2x+1)$$

$$y' = \frac{2x+1}{2\sqrt{x^2 + x - 2}} = 0 \quad \text{No hay}$$

$$y'' = \frac{2 \cdot 2\sqrt{x^2 + x - 2} - (2x+1) \cdot \frac{1}{\sqrt{x^2 + x - 2}}}{(2\sqrt{x^2 + x - 2})^2} \cdot (2x+1)$$

$$y'' = \frac{1}{\sqrt{x^2 + x - 2} (4x^2 + 4 - 8)} = \text{no hay}$$

37) $y = \sin x + \sqrt{3} \cos x$

$$-2\pi \leq x \leq 2\pi$$

$$y' = \cos x - \sqrt{3} \sin x$$

$$-\sqrt{3} \sin x = -\cos x$$

$$-\sqrt{3} \tan x = -1$$

$$\tan x = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$$

$$x = \tan^{-1} \frac{\sqrt{3}}{3} = \frac{\pi}{6}$$

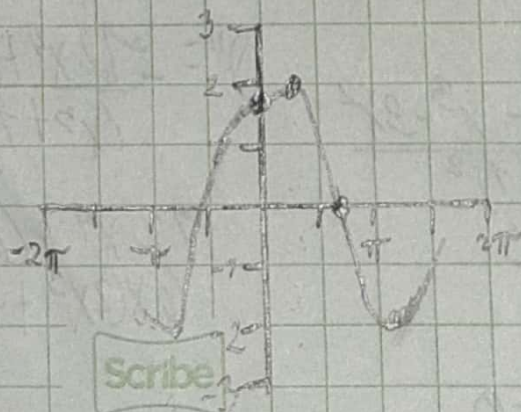
$$y' = -\sqrt{3} \cos x - \sin x = 0$$

$$-\sin x = \sqrt{3} \cos x$$

$$-\tan x = \sqrt{3}$$

$$x = \tan^{-1} -\sqrt{3} = -\frac{\pi}{3} = -\frac{2\pi}{6}$$

	$f'(x)$	$f''(x)$	$f(x)$
$-2\pi, \pi/6$	-	+	
$\pi/6$			2
$\pi/6, 2\pi/3$	-	-	
$2\pi/3$			0
$2\pi/3, 2\pi$	-	+	



$$39) y = \frac{\sin x}{1 + \cos x}$$

$$y' = \tan\left(\frac{x}{2}\right)$$

$$y = \sec\left(\frac{x}{2}\right) \neq \frac{1}{2}$$

$$y' = \frac{\sec\left(\frac{x}{2}\right)^2}{2} = 0$$

$$\sec\left(\frac{x}{2}\right) = 0$$

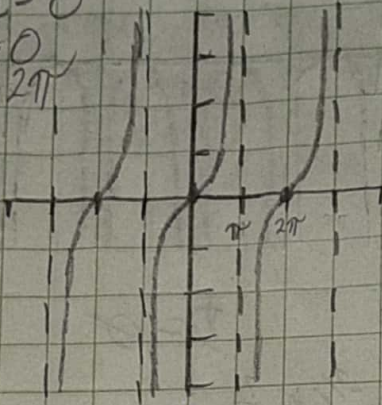
$$y'' = \frac{1}{2} + 2\sec\left(\frac{x}{2}\right)\tan\left(\frac{x}{2}\right)\sec\left(\frac{x}{2}\right)\frac{1}{2}$$

$$y'' = \frac{\sec^3 \frac{x}{2}}{2} = 0$$

$$\sec \frac{x}{2} = 0$$

$$\frac{x}{2} = 0$$

$$x = 2\pi$$



	$f'(x)$	$f(x)$
$-\infty, 2\pi$	+	
2π		0
$2\pi, \infty$	+	

$$41) y = \arctan(e^x)$$

$$y' = \frac{1}{1 + (e^x)^2} \neq e^x$$

$$y' = \frac{e^x}{1 + e^{2x}} = 0$$

$$y'' = \frac{e^x(1 + e^{2x}) - e^x e^{2x} \cdot 2}{(1 + e^{2x})^2}$$

$$y'' = \frac{e^x - e^{3x}}{(1 + e^{2x})^2} = 0$$

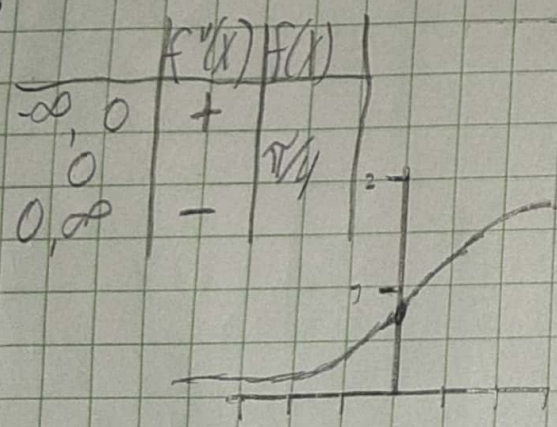
$$e^x - e^{3x} = 0$$

$$e^x = e^{3x}$$

$$x = 3x$$

$$2x = 0$$

$$x = 0$$



Scribe

$$49) y = \ln(\sin x)$$

$$y' = \frac{1}{\sin x} \cdot \cos x$$

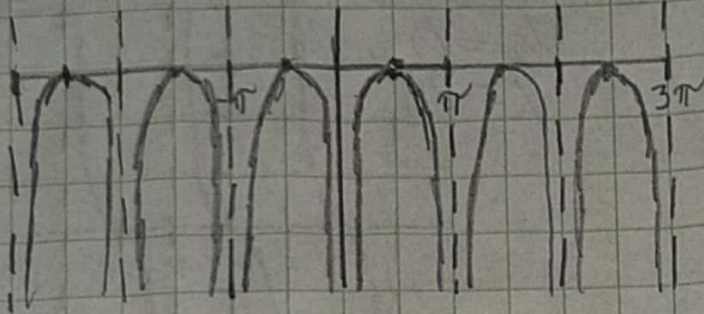
$$y' = \cot x = 0$$

$$x = \pi/2$$

$$y'' = -\csc^2 x = 0$$

	$f'(x)$	$f(x)$
$-\infty, \pi/2$	+	
$\pi/2$		0
$\pi/2, \infty$	-	

$$\text{domain} = \pi, 2\pi, \dots$$



$$51) y = xe^{-1/x}$$

$$D = \emptyset$$

$$y' = e^{-1/x} + xe^{-1/x} \cdot \frac{1}{x^2}$$

$$y' = \frac{1}{e^{1/x}} + \frac{1}{xe^{1/x}} = 0$$

$$\frac{x+1}{e^{1/x} x} = 0$$

$$x+1=0$$

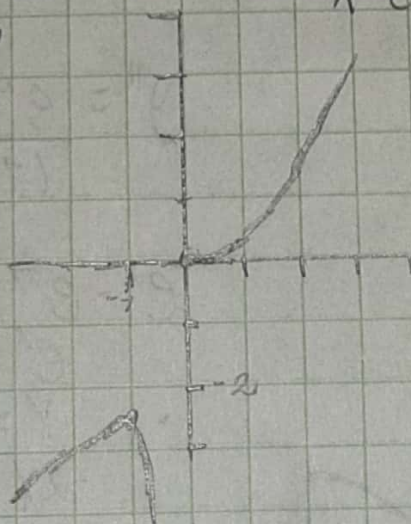
$$x=-1$$

	$f'(x)$	$f(x)$
$-\infty, -1$	+	
-1		-2.72
$-1, \infty$	-	

$$y'' = \frac{-e^{1/x}(-\frac{1}{x^2}) - e^{1/x} + xe^{1/x}(\frac{1}{x^2})}{(e^{1/x})^2}$$

$$y'' = \frac{1}{x^3 e^{1/x}} = 0$$

$$1 = 0$$

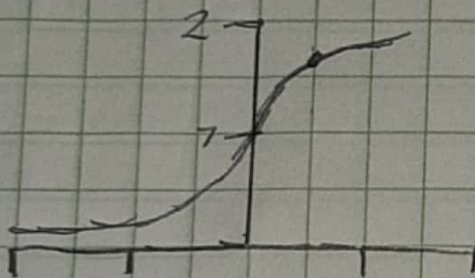


$$53) y = e^{\tan^{-1}x}$$

$$y' = e^{\tan^{-1}x} \cdot \frac{1}{1+x^2}$$

$$y' = \frac{e^{\tan^{-1}x}}{1+x^2} = 0$$

	$f''(x)$	$f(x)$
$-\infty, 1/2$	+	
$1/2$		1.6
$1/2, \infty$	-	



$$y'' = \frac{e^{\tan^{-1}x} \left(\frac{1}{1+x^2} \right) (1+x^2) - e^{\tan^{-1}x} (2x)}{(1+x^2)^2}$$

$$y'' = \frac{e^{\tan^{-1}x} - 2e^{\tan^{-1}x}x}{(1+x^2)^2} = 0$$

$$e^{\tan^{-1}x} - 2e^{\tan^{-1}x}x = 0$$

$$e^{\tan^{-1}x} (1 - 2x) = 0$$

$$1 - 2x = 0$$

$$x = \frac{1}{2}$$